

2024

A Division of Ventacity Systems



UBX Systems Delivers a Smart Building System That’s up to 20x More Cost Effective Using AWS IoT

Learn how Ventacity Systems, a manufacturer of smart building technology, developed UBX, an IoT solution using AWS and LoRaWAN devices.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Up to 20x

more cost efficient than wired smart b



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



50x less



infrastructure costs than wireless LAN or wireless mesh networks

130,000+

square feet area covered by a single LoRaWAN hub

Over 10x faster

install than traditional wired solutions

Overview

Heating and cooling a commercial building require a lot of energy. According to a [report](#) by the US Energy Information Administration, commercial buildings consumed 18 percent of all energy in the United States in 2022. Although smart systems have helped many businesses optimize their energy use, installation of these systems is challenging. It can require significant installation time and effort. Moreover, deploying smart devices with wired connectivity in a building can cost hundreds of thousands of dollars.

Manufacturing company [Ventacity Systems](#) (Ventacity) wanted to help more customers achieve their sustainability goals by lowering the barrier to entry for smart building technology. So Ventacity developed the [Unified Building Experience](#) (UBX) solution on Amazon Web Services (AWS) and created the UBX Systems division to oversee it. To establish secure and reliable cloud connectivity across its customers' buildings, UBX uses [AWS IoT Core for LoRaWAN](#), which can connect, manage, and secure LoRaWAN devices at scale without requiring customers to develop, maintain, or operate their own LoRaWAN infrastructure.



**Opportunity | Reducing Smart Building Complexity with
AWS IoT Core for LoRaWAN**



UBX Systems develops smart solutions that provide remote monitoring, analysis, and control of HVAC systems, as well as nearly any LoRaWAN device, from smart thermostats to occupancy and energy monitoring solutions.

The first iteration of the UBX smart building solution connected its customers' HVAC systems and remote sensors to the internet with a Linux-based controller. However, the controllers and sensors required wired connections within the building, which could take weeks to install. "We wanted to go entirely wireless and cover entire buildings without pulling any cables," says Jonah Peskin, chief technology officer for UBX Systems.

The UBX Systems team evaluated its options, including Bluetooth, wireless LAN, and various advanced wireless mesh networks. The drawbacks were that customers would still need to install multiple gateways, hubs, or repeaters with wired power sources as well as potentially thousands of nodes throughout their building to achieve full coverage.

The company then considered LoRaWAN connectivity, a low-power, wide-area networking protocol designed to support wireless connectivity for low-power IoT devices over long distances. The UBX team learned about AWS IoT Core for LoRaWAN when it became generally available. "We can go into a seven-story building that's 130,000 square feet and connect every device with a single wireless hub," says Peskin. "AWS IoT Core for LoRaWAN solved all our in-building connectivity problems. You no longer need to deploy and manage multiple gateways; it's revolutionary."

Solution | Achieving 20x Lower Costs Than Wired Systems with an All-Wireless Smart Building System

UBX Systems developed a LoRaWAN-based system with a hub, which can be plugged into any 120-volt power outlet. Customers can then monitor their environments and control their equipment by placing small, battery-powered LoRaWAN sensors or devices throughout their buildings. The cloud software automatically processes and segments IoT data using [AWS Lambda](#), a compute service that runs code in response to events and automatically manages compute resources, to run code without clusters or servers.

A single hub can scale to support thousands of nodes at a time. UBX also offers the option to add more hubs to increase both coverage area and reliability, where IoT data transmissions traverse every available hub and are then de-duplicated in the cloud on AWS. "Each hub, powered by AWS IoT Core for LoRaWAN, routes data to AWS, where the fully managed LoRaWAN Network Server lives," says Peskin. "That gives us a great user experience. Any of our customers' gateways could be unplugged accidentally, and they would not lose data. It's also very simple to swap out and add new hubs for even more wireless coverage and robustness." UBX stores



customers' device encryption keys in the cloud instead of using on-premises servers, which helps to protect their devices from malicious threats.

Compared with wired smart building solutions, the UBX solution is 6–20 times more cost effective. Because it is based on LoRaWAN protocol and directly connected to the cloud through AWS IoT Core for LoRaWAN, it requires minimal maintenance. In addition, “LoRaWAN devices are unprecedented in their battery usage,” says Peskin. “We’re getting real-world examples of devices that can run for 15 years on a single set of batteries.”

UBX customers can access smart insights through a web-based application, the UBX Application Suite. From there, they can improve the comfort and efficiency of their buildings by monitoring elements such as temperature, humidity, CO2, occupancy, equipment status, and energy in near real time to better understand patterns in their building usage and energy consumption. “People are more concerned than ever about comfortable, healthy buildings and energy use, and simply understanding what is going on in your facility is an essential first step,” says Peskin.

The UBX Application Suite also provides customers the opportunity to build a digital twin or virtual building replica, set up preventive maintenance alerts, and remotely control their HVAC systems and smart thermostats. Plus, UBX offers unique LoRaWAN adapters to bridge traditional wired building protocols (Modbus and BACnet) to the wireless LoRaWAN network. “We make it possible to add virtually any existing HVAC or building subsystem to a UBX smart building,” says Peskin.

Outcome | Developing UBX to Support a Wide Variety of Use Cases and Devices

UBX Systems is scaling its deployments and has installed its solution in a broad range of buildings, including offices, schools, retail, restaurants, theaters, medical facilities, and more. To support this growth, UBX Systems must efficiently manage devices across geographies. Using AWS IoT Core for LoRaWAN, UBX Systems can scale its device deployments globally, keep a pulse on operations, and perform targeted over-the-air updates across its fleet.

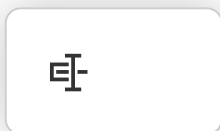
Despite the uptick in demand, the company has kept its operations running smoothly, reacting quickly to unforeseen issues while reducing its cloud costs by five times. “In addition to cost savings, using AWS IoT Core for LoRaWAN for UBX makes it simple to add data points, devices, and new applications, such as smart thermostats, comfort and air quality monitoring, refrigeration monitoring, energy monitoring, and equipment monitoring and control with very little added infrastructure,” says Peskin.

UBX also uses the broad ecosystem of LoRaWAN vendors to deploy solutions for almost any business need. “If we want to add a water leak sensor or gas metering, it’s quick and simple,” says Peskin. “We can solve just about any smart building challenge by combining LoRaWAN and our UBX technologies on AWS.”

About Ventacity Systems

Ventacity’s vision is to make buildings healthy, efficient, and smart. Its UBX solution is a wireless smart building solution providing customers with the ability to monitor and control all aspects of a building’s environment and equipment.

AWS Services Used



AWS IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)

AWS IoT Core for LoRaWAN

AWS IoT Core for LoRaWAN is a fully managed LoRaWAN Network Server (LNS) that enables customers to connect wireless devices that use the LoRaWAN protocol for low-power, long-range wide area network connectivity with the AWS Cloud.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

More Financial Services Customer Stories

HONG KONG



Bitlayer Builds the First Layer 2 BitVM Platform...

Learn how Bitlayer created the world's first BitVM-based Layer 2 solution for

HONG KONG



The Bank of East Asia Improves Its Mobile App...

Learn how The Bank of East Asia, Limited (BEA) delivers a better mobile



Bitcoin, processing over 150,000 transactions daily and supporting over 300 projects using Amazon EC2.

2024

app experience on AWS, helping accelerate its omnichannel strategy to provide exceptional customer experiences.

2024

EUROPE, MIDDLE EAST, & AFRICA



TP ICAP on AWS

Liquidity and data solutions specialist TP ICAP is trusted by clients worldwide for its market intelligence, data and analytics, and broking services. The company has moved 50 percent of its IT estate to Amazon Web Services, giving it the scale and security to run critical financial trading platforms as well as accelerating decarbonization and supporting its commitment to sustainability. By using the latest generative artificial intelligence



2024

UNITED STATES



BankUnited Enhances Customer and Employee...

Learn how BankUnited, a financial services institution, built a generative AI solution on AWS to transform its operations and improve customer support.

2024

1 2 3 4 5 6 ... 122

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales



Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |
- Tiếng Việt |
- Türkçe |
- Русский |
- ไทย |
- 日本語 |
- 한국어 |

2

